

# A Regime-Aware Model for Loan-Level Probability of Default

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# Chapter 1

## Introduction

This thesis studies whether macro regime information can improve loan-level probability of default estimation for residential mortgages in a way that remains interpretable, operationally useful, and calibration-focused. The active empirical setting uses Freddie Mac sample mortgage data together with monthly macroeconomic series from FRED.

### 1.1 Problem Statement

Loan-level mortgage PD models are often estimated as if borrower risk relationships remain stable through time. That simplification becomes weak when inflation, monetary tightening, and labor-market stress alter repayment behavior. A pooled model may still rank risk reasonably well while producing poorly calibrated probabilities when macro conditions shift.

### 1.2 Empirical Thesis Direction

The thesis has been reset away from a simulation-first framing and toward an empirical mortgage study. The active research design combines two layers:

- a macro regime layer, where a hidden Markov model is fit to monthly inflation, interest-rate, and unemployment data, and
- a loan-level PD layer, where the inferred stress probability is introduced into an interpretable logistic model through borrower-feature interactions.

The central question is whether this regime-aware logistic specification improves probability quality relative to simpler benchmarks when evaluated on mortgage panel data.

### 1.3 Contribution Claim

The contribution is not framed as a general claim that regime switching is new in credit risk. The narrower contribution is an interpretable mortgage PD framework that:

- infers latent macro stress from public macroeconomic data,
- carries that signal into a logistic PD model through a transparent interaction structure,
- compares the resulting model with borrower-only and macro-augmented logistic baselines, and
- evaluates success primarily through calibration-oriented metrics rather than prediction accuracy alone.

This positioning deliberately separates the thesis from black-box forecasting work such as Brookfield’s gradient boosting approach. Brookfield remains related work, not the empirical benchmark being implemented here.

## 1.4 Data Setting

The active data plan uses Freddie Mac sample origination and servicing files for cohorts from 2015 through 2025, combined with monthly FRED series for inflation, the Federal Funds rate, and unemployment. The unit of analysis is a loan-month observation. The main target is a forward-looking serious-delinquency indicator defined over a 12-month horizon.

## 1.5 Current Repository Status

At the time of this reset, the repository documents the empirical direction but does not yet implement it end to end. The existing codebase remains a legacy synthetic scaffold. Synthetic notebooks and generated outputs have been archived so that the active documents no longer present them as current thesis evidence.

# Chapter 2

## Literature Review

The literature relevant to this thesis sits at the intersection of loan-level PD modeling, macro-sensitive credit risk analysis, regime-switching methods, and calibration-focused model assessment.

### 2.1 Loan-Level PD Modeling

Logistic regression remains a standard benchmark for default modeling because it is interpretable, easy to audit, and naturally aligned with probability estimation. In mortgage settings, borrower and loan characteristics such as origination leverage, credit quality, product structure, delinquency status, and loan age often provide the core explanatory signal.

### 2.2 Macro-Augmented Credit Risk Models

Another branch of the literature adds observed macro variables directly into credit risk models. This is useful for stress testing and cyclical interpretation, but it still assumes that macro conditions affect default probabilities through a fixed functional form. When macro conditions change sharply, a model with static coefficients may miss regime-dependent borrower sensitivities.

### 2.3 Regime-Switching Approaches

Regime-switching and hidden-state models offer a way to summarize broad economic conditions as latent states rather than as a fixed linear macro effect. In this thesis, the hidden Markov model is not the end product. It is a macro-state extraction layer that produces a filtered stress probability for downstream PD modeling.

### 2.4 Positioning Relative To Brookfield

Brookfield is the closest recent related work in spirit because it treats macro sensitivity and regime behavior as important for consumer credit forecasting. However, the present thesis

does not attempt to replicate Brookfield’s gradient boosting architecture. Brookfield is used as a literature reference that motivates the importance of regime sensitivity. The active contribution here is narrower and more interpretable: a regime-aware logistic PD model for mortgage data.

## **2.5 Gap Addressed By This Thesis**

The gap targeted by this thesis is practical and methodological rather than purely predictive. The study asks whether a latent macro stress signal can improve loan-level mortgage PD estimation while preserving transparency, defensible model comparison, and calibration-focused evaluation. That is the basis for the empirical design described in the next chapter.

# Chapter 3

## Methodology

### 3.1 Data Sources

The planned empirical workflow combines two public data sources:

- Freddie Mac sample origination and servicing files for mortgage cohorts from 2015 through 2025
- FRED monthly macroeconomic data for inflation, the Federal Funds rate, and unemployment

The mortgage data provides the loan-level panel structure, while the macro data supplies the aggregate environment used for regime inference.

### 3.2 Unit Of Analysis

The unit of observation is one mortgage at one month. Each loan-month observation is intended to combine:

- origination characteristics,
- allowed current performance indicators from servicing records,
- month-level macro variables, and
- a forward-looking default proxy label.

### 3.3 Primary Label

The active label plan is a 12-month transition to 90+ days past due. For a loan-month indexed by  $(i, t)$ , the target equals one when the loan is not already 90+ DPD at month  $t$  and reaches serious delinquency within the next 12 months. Observations without a full forward horizon will be excluded from model training and evaluation.

### 3.4 Macro Regime Inference

Let  $M_t$  denote the monthly macro vector containing inflation, the Federal Funds rate, and unemployment. A two-state Gaussian hidden Markov model will be fit to the standardized macro series:

$$M_t \mid R_t = r \sim \mathcal{N}(\mu_r, \Sigma_r),$$

where  $R_t \in \{1, 2\}$  is the latent macro regime. The main downstream quantity is the filtered stress probability

$$p_{\text{stress}}(t) = \Pr(R_t = \text{stress} \mid M_{1:t}).$$

Filtered rather than smoothed probabilities are planned so that the downstream PD model does not depend on future macro information.

### 3.5 Planned Model Comparison

The active empirical study will compare exactly three logistic specifications:

1. borrower-only baseline
2. macro-augmented baseline
3. regime-aware logit with  $p_{\text{stress}}(t)$  and borrower-feature interactions

The regime-aware specification is planned as

$$\Pr(Y_{i,t}^{(12)} = 1 \mid x_{i,t}, p_{\text{stress}}(t)) = \sigma(\beta_0 + \beta^\top x_{i,t} + \gamma p_{\text{stress}}(t) + p_{\text{stress}}(t) \theta^\top x_{i,t}).$$

This keeps the final model interpretable while allowing borrower risk sensitivities to vary with macro stress intensity.

### 3.6 Evaluation Plan

Evaluation is calibration-first. The primary performance tools will be:

- Brier score
- calibration plots
- calibration by regime slice and time slice

Secondary metrics will include ROC-AUC, PR-AUC, and log loss. All comparisons are intended to use chronological splits only.

## 3.7 Implementation Status

The repository has been reset to document this methodology, but the empirical code migration has not yet been completed. The current code remains legacy synthetic scaffolding pending replacement by a Freddie Mac ingestion and empirical modeling workflow.

# Chapter 4

## Results

This chapter is currently a results plan rather than a report of completed empirical findings. The repository has been reset before the empirical pipeline has been rebuilt, so the active manuscript now documents how results will be presented once the Freddie Mac workflow is implemented.

### 4.1 Planned Results Structure

The final empirical results chapter should report four layers of evidence:

- descriptive summaries of the mortgage panel and macro sample overlap
- the inferred macro regime path and stress probabilities from the HMM
- comparative model results for the borrower-only, macro-augmented, and regime-aware logistic models
- calibration-focused diagnostics by time period and macro regime

### 4.2 Planned Tables And Figures

The intended output set includes:

- a data coverage table for Freddie Mac cohorts and usable loan-month observations
- a figure showing the macro regime path and filtered stress probability
- a model comparison table with Brier score, ROC-AUC, PR-AUC, and log loss
- calibration plots for each model on the held-out period
- regime-slice and time-slice evaluation tables

### **4.3 Interpretation Focus**

The chapter will not be organized around prediction accuracy alone. The main interpretation goal is to determine whether the regime-aware logit produces more reliable default probabilities, especially during macro stress and transition periods.

### **4.4 Current Reset Status**

Synthetic figures and synthetic performance tables from the earlier repository phase have been archived and are not part of the active thesis evidence. New active results will only be added after the empirical pipeline has been implemented and validated.

# Chapter 5

## Conclusion

The thesis has been reset toward an empirical Freddie Mac plus FRED study of regime-aware mortgage PD modeling. The active contribution is now framed as an interpretable logistic framework that uses an HMM-derived macro stress probability to capture changing borrower sensitivities across macro conditions.

### 5.1 Expected Contribution

If the empirical workflow performs as planned, the thesis will contribute a clear comparison between borrower-only, macro-augmented, and regime-aware logistic PD models in a mortgage setting, with calibration treated as a primary outcome rather than an afterthought.

### 5.2 Main Limitation At This Stage

The repository is still in transition. The narrative and documentation have been updated first, while the implementation remains to be migrated from the archived synthetic scaffold to the Freddie Mac workflow.

### 5.3 Next Steps

The next implementation steps are:

- Freddie Mac origination and servicing ingestion
- loan-month panel construction and label definition
- macro HMM estimation using FRED inflation, Fed Funds, and unemployment
- three-model logistic comparison with chronological validation
- calibration-first results reporting

That sequence defines the active path from repository reset to empirical execution.

# Bibliography